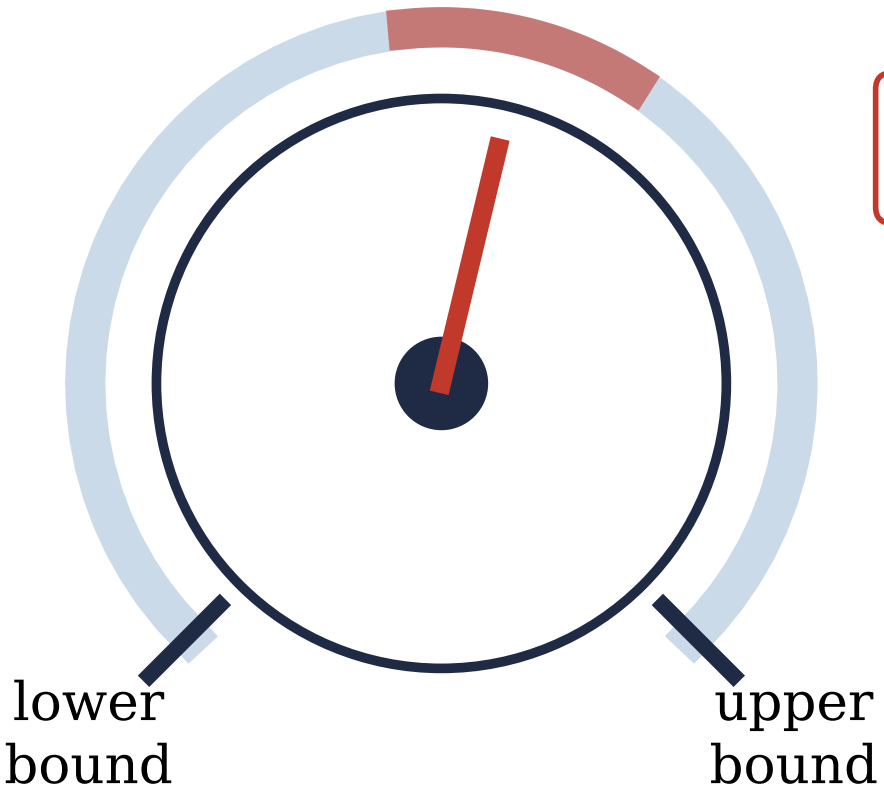

the prior: every setting allowed before looking (blue arc)

the data-allowed width: settings the data still permit (red arc)

the setting: the best value (the pointer)

bits = $\log_2$(blue / red): how much the data taught you